1. N83 spike38 exact original-top degree of regularity

Run: `n83-chain-exact-spike38-20261111-r1` **First computation:** 2026-11-11 02:58 UTC **Study:** fixed factor-base index-calculus equations, source-bound exact census

1.1. Result

The exact original-individual-top degree of regularity was certified for all 28 frozen cases: Koblitz and general binary families, m=3 through 9, with natural and planted targets. All four cases at each m have the same degree. The values are 111, 194, 277, 360, 443, 526, and 609, respectively. They equal the frozen prediction `83(m-2)+28` at every measured m.

m	Exact dreg	Predicted	Cases
3	111	111	4
4	194	194	4
5	277	277	4
6	360	360	4
7	443	443	4
8	526	526	4
9	609	609	4

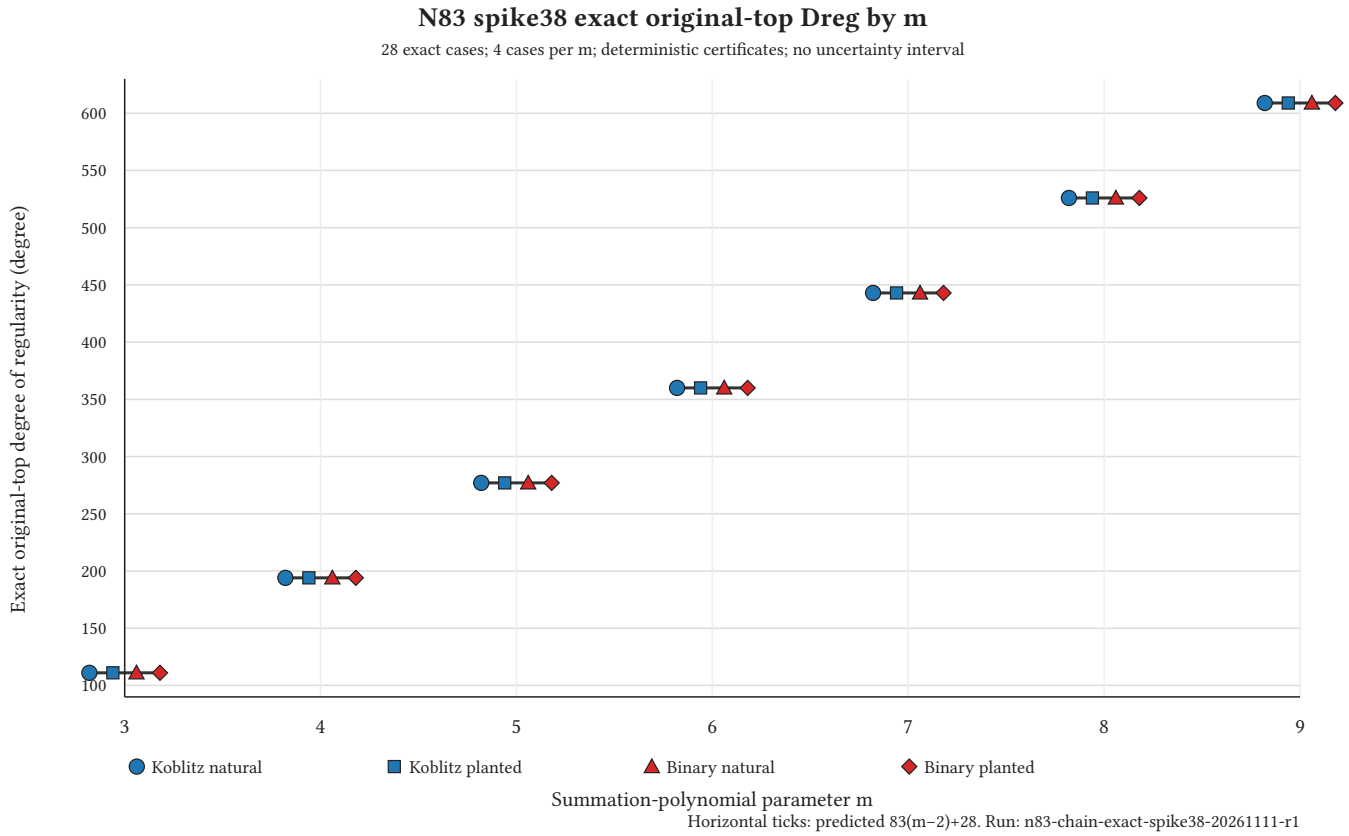


Figure 1: Exact values from the 28 certified run rows; ticks show the frozen formula. Each plotted point is one case. Run `n83-chain-exact-spike38-20261111-r1`.

1.2. What was measured

The primary metric is the first degree where the homogeneous quotient is zero for the ideal generated by the original individual highest forms, including Boolean squares. Affine row combinations were not taken before forming the highest forms. The result is a degree-of-regularity census for the frozen equation systems and factor-base inputs. It is not first-fall degree, last-fall degree, affine solving degree, or a timed attack-cost measurement.

The frozen panel contains 28 unique inputs: two families x seven m values x natural/planted target. The inputs and source files are SHA-256 bound in `inputs.manifest.json` and retain their existing S3 object URIs. The selected factor-base manifest is `r=38`, with hashes recorded per case. No factor-base points or targets were sampled during this run.

1.3. Checks and execution

- Producer certificates: 28 rows, `all_verified=true`, zero errors.
- Fresh-process replay: `all_passed=true`, 28 cases; replay summary hash `c042307355c90cdd91c7bc4ed91ec98880a9d58fb32b6164d64a415c9e6c887a`.
- Planted positive arm: all planted cases passed the same source parser and certificate path.
- A/A check: the replay used the same binary and exact same inputs in a separate process; all rows matched.
- Negative n83 arm: not applicable because this absolute census has no matched input outside the theorem hypotheses. Small structural positive/negative controls are retained in `certificates/controls.json` and are not presented as an n83 null arm.
- Frozen protocol digest: `ff565b060fb916e19f12cc168d1def9166f088de117f24aa8853a65057cba43f`.
- Producer/replay binary SHA-256: `0f8b269e44a2c154de6db4979e0bd99f7f3236f91dce5820de705c0a144ba951`; source file SHA-256: `ecba2c205ea3e2a04622ee2be3b6d94c7ee489f4390ac095f3b20fb4deaa8d85`.
- Producer elapsed time 35.86 s, replay 34.56 s; peak RSS 142,864 KiB and 144,116 KiB, respectively. These are L0 process diagnostics, not performance evidence.
- The producer wrapper had a typo only in its final exit-marker path. `/usr/bin/time` recorded producer exit status 0; the correct `producer.exit` and the wrapper issue are documented in the run directory.

The run bundle has 76 manifest-listed files totaling 21,175,275 bytes. Its SHA-256 manifest was verified locally. After the push, the repository's artifact-autofix workflow moved the 9,276,061-byte `certificates/global.json` into the content-addressed S3 store and committed the pointer at `certificates/.artifacts.json`. I downloaded that object and verified its SHA-256 against the manifest: `s3://setec-astronomy-artifacts/store/sha256/16/168f16028492ac47c3f35e16937d67c504bce635278592955fdb07c5e79675de`. The configured monthly budget readback was already USD 1,360.114 against a USD 1,300 limit; the workflow did not expose an attributable per-object cost. No EKS job was submitted.

1.4. Scope and next work

The proved formula's stated scope is the Koblitz/binary chain with m>=3 and the frozen selector hypotheses. This run therefore does not add prime-field or m=2 measurements. It also does not calculate FFD or LFD, rank factor bases, or establish a general solving-time consequence. The native certificate bridge has not yet received a full Lean formalization or independent scientific review.

The next measurement is a separately frozen extension for the remaining families and m values, using the same immutable point bases and an appropriate method for inputs outside this theorem's hypotheses. FFD/LFD labels and grouped model training remain separate downstream work.

1.5. References

- Kosters and Yeo, **Notes on summation polynomials**, for characteristic-two summation-polynomial context and the distinction between first fall and original-top regularity.
- Caminata and Gorla, **Solving degree, last fall degree, and related invariants**, Definitions 1.3–1.5.
- Li and Kadir, Moore matrix background, used only for the stated matrix context; the run checks its specific inverses directly.

Machine evidence: `evidence/chain-exact-spike38-20261111/` **Plot source:** `scripts/plot_chain_exact_dreg_svg.py`; plot data: `evidence/chain-exact-spike38-20261111/plot/plot-data.csv`.